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Book No. 1 (containing 28 pages)

# THE INSTITUTE OF CHARTERED ACCOUNTANTS OF INDIA

Final Examination

Group No 1 Paper No 2

Subject AFM

Number of Answer Books used : Main + 5 additional sheets

Date Seal

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818322

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF INDIA SEAL MAY, 2025

04 MAY 2025

| Paper Code | |
|---|---|
| L | (L) |
| M | (M) |
| N | (N) |
| 1 | (1) |

| MCQ Booklet Serial No. | Paper No. | Level of Exam: Final |
|---|---|---|
| 5122913 | 2 | Foundation (1) Intermediate (2) Final (●) |

MCQ Answers

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
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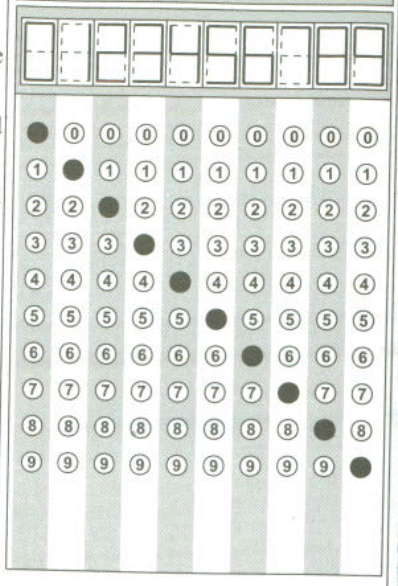
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| 2 | ☐ | 9 | ☐ |
| 3 | ☐ | 10 | ☐ |
| 4 | ☐ | 11 | ☐ |
| 5 | ☐ | 12 | ☐ |
| 6 | ☐ | 13 | ☐ |
| 7 | ☐ | 14 | ☐ |
| Total | | Total | |

## Answer - 5(a)

Q.5.

a.

(i) Income Available for Distribution

| | Units (lakhs) | Income per Unit | Total Income (₹ lakh) |
|---|---|---|---|
| Income of April | 800 | 0.0618 | 49.440 |
| Issue of ~~Shares~~ Units | 16 | 0.0618 | 0.9888 |
| | 816 | 0.0618 | 50.4288 |
| Income of May | 816 | 0.0920 | 75.070 |
| | 816 | 0.1538 | 125.4988 |
| ~~Income of May~~ | ~~816~~ | ~~0.0920~~ | ~~75.070~~ (1.2304) |
| Repurchase | (8) | 0.1538 ~~0.0920~~ | ~~(0.736)~~ |
| | ~~808~~ | ~~0.0920~~ | ~~74.334~~ |
| | 808 | 0.1538 | 124.2684 |
| Income of June | 808 | 0.1160 | 93.710 |
| | 808 | ~~0.2080~~ | ~~168.044~~ |
| | | 0.2698 | 217.9784 |
| ~~(-) 60% Distributed~~ | ~~808~~ | ~~0.1248~~ | ~~(100.8264)~~ |
| | ~~808~~ | ~~0.0832~~ | ~~67.2176~~ |
| (-) 60% Distributed | 808 | 0.1619 | (130.7870) |
| | 808 | 0.1079 | 87.1914 |

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(ii) Issue Price for April

| | ₹ |
|---|---|
| NAV | 44.30 |
| (+) 2% Load | 0.886 |
| (+) Dividend Equalization | 0.0618 |
| | 45.2478 |

(iii) Repurchase Price for May

| | | ₹ |
|---|---|---|
| NAV | | 44.30 |
| (-) 2% Load | | (0.886) |
| (+) Dividend Equilazation | 0.1538 | ~~0.0420~~ |
| | | ~~43.506~~ |
| | | 43.5678 |

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(iv) NAV on 30th June

| | | (₹ lakhs) |
|---|---|---|
| Opening Net Assets (800 Lakhs × ₹ 44.30) | | 35,440 |
| (+) Portfolio Appreciation | | 965.280 |
| (+) Income of April | | 49.440 |
| (+) Income of May | | 75.070 |
| (+) Income of June | | 93.710 |
| (+) Receipts on Issue (45.2478 × 16 lakhs) | | 723.9648 |
| (-) Payment on Repurchase (~~43.506~~ 43.5678 × 8 Lakhs) | | ~~(348.048)~~ (348.5424) |
| (-) Dividend | (130.7870) | ~~(67.2176)~~ |
| | 36868.1354 | ~~36932.1992~~ |
| Units | | 808 lakhs |
| NAV | 45.6289 | ~~45.7082~~ |
| | OR | ~~or~~ |
| | 45.63 | ~~45.71~~ |

## Answer - 5(b)

Q.5.
b.

(A) Value Based on EV/EBITDA

| | |
|---|---|
| EBITDA = ₹ ~~100~~ 110 lakhs | |
| (-) Extraordinary Gain | (₹16 lakh) |
| (-) Sales Promotion | (₹38 lakh) |
| | ₹56 lakhs |
| EV/EBITDA Multiple | 8 |
| ∴ Equity Value | ₹448 lakhs |

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(B) Value based on DCF Model.

(i) Computation of Cost of Equity

Unlevered Beta = 1.5

$$\text{Levered Beta} = \text{Unlevered Beta}\left[\frac{\text{Debt + Equity (Post Tax)}}{\text{Equity}}\right]$$

$$= 1.5\left[\frac{(0.3)(1-0.3)+0.7}{0.7}\right]$$

$$= 1.5\left(\frac{0.91}{0.7}\right)$$

$$= 1.95$$

$$Ke = R_F + \beta(R_m - R_F)$$

$$= 6 + 1.95(12-6)$$

$$= 17.7\%.$$

(ii) Computation of WACC

$$WACC = (ke \times weight) + (kd\ (post\ tax) \times w)$$

$$= 17.7\,(0.7) + 12.45\,(1-0.3)\,(0.3)$$

$$= 15.0045\%$$

$$\approx \boxed{15\%}$$

(iii)

| Year | Cash Flows | Present Value Factor | Present Value |
|---|---|---|---|
| 1 | 150 | 0.870 | 130.5 |
| 2 | 200 | 0.756 | 151.2 |
| 3 | 220 | 0.658 | 144.76 |
| | | | 426.46 |

∴ Value = ₹426.46 Lakhs.

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## Answer - 6 (a)

Q.6.
a

Contract Value = 2 lot x ₹ 25,400 x 50 units

= ₹ 25,40,000

Initial Margin = 25,40,000 x 10%.

= 254,000

Maintenance Margin = 254,000 x 80%.

= 203,200.

(i) Daily Margin Balance

| Date | Particulars | Balance | Payment |
|---|---|---|---|
| - | - | 254,000 | - |
| 1 Jan | (25,520 - 25400) x 50 x 2 = 12000 | 266,000 | - |
| 2 Jan | (25390 - 25520) x 50 x 2 = (13000) | 253,000 | - |
| 3 Jan | (25250 - 25390) x 50 x 2 = (14000) | 239000 | - |
| 4 Jan | (24800 - 25250) x 50 x 2 = (45000) | 254000 | 60,000 * |
| 5 Jan | (25100 - 24800) x 50 x 2 = 30,000 | 284,000 | - |

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* Call Money Paid = ~~(239000 - 45000)~~

$254000 - (239000 - 45000)$

$254000 - 194000$

$= \boxed{₹60,000}$.

(ii) Gain / Loss on Square Off

| | ₹ |
|---|---|
| Contract Bought at | 25,400 |
| Sold at | 25,100 |
| | (300) |

∴ Loss = ~~₹400 × 50 × 2~~

= ~~₹40,000 Loss~~

₹300 × 50 × 2

30,000 Loss

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(iii) Gain / Loss on Short Position

| | ₹ |
|---|---|
| Contract Sold at | 25,400 |
| Bought at | 25,100 |
| | 300 |

∴ Gain = 300 × 50 unit × 2 lots

= **₹ 30,000 Gain**

## Answer - 6(b)

Q.6.

b.

(i) Cap Rate = 8%.
Premium = 0.75%.
Current Rate = 10%.

It is assumed that cap option has been purchased for period of 1 year.

∴ Amount Received = Notional Principal × [Current Rate − Cap Rate]

= 50,00,000 × (10% − 8%)

= $ 100,000.

Cost of Cap = 50,00,000 × 0.75%.

= $ 37500

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$\therefore$ Net Savings = 100,000 - 37500

= $\boxed{\$62,500}$.

(ii)

DEF Ltd also purchases Floor @ 5%.

∴ Premium on Floor = 50,00,000 × 0.85%.

= $42,500

Amount received on Cap = $100,000

Cost of Cap = $37,500

Amount received on Floor = Nil, as Floor will not be exercised

∴ Net Saving = 100,000 − 37500 − 42500

= $20,000.

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(iii) Interest Rates fall to 4%.

Amount Received on Cap = Nil, as not exercised

Amound Received on Floor = ~~50,00,000~~

Notional Principal × [Floor Rate (–) Market Interest Rate]

= 50,00,000 (5% - 4%)

= $50,000

Cost of Cap = $37,500
Cost of Floor = $42,500

∴ Net Saving = 50,000 - 37500 - 42500

= (30,000)

∴ Net Cost = $30,000

(iv)

Purchases Cap and Sells Floor

Amount Received under Cap = 50,00,000 (11% - 8%)

= $150,000

Amount Raid under Floor = Nil, as not exercised

Cost of Cap = $ 37,500

Premium Received on Floor = $ 42,500

Net Saving = 150,000 - 37,500 + 42,500

= $155,000

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## Answer- 2(a)

Q.2.

a

| Security | $\sigma$ | Weight |
|---|---|---|
| R | 20% | 0.8 |
| S | 50% | 0.2 |

(1) Returns are independent

$\therefore r = 0$

$$\therefore \sigma_p = \sqrt{\sigma_R^2 w_R^2 + \sigma_S^2 + w_S^2 + 2\sigma_R \sigma_S w_R w_S r_{RS}}$$

$$= \sqrt{(0.2)^2(0.8)^2 + (0.5)^2(0.2)^2 + 0}$$

$$= \sqrt{0.0256 + 0.01}$$

$$\sigma_p = 0.1887$$

$$= \boxed{18.87\%}$$

(ii) Returns are perfectly negatively correlated

$$\sigma_p = \sqrt{\sigma_R^2 w_R^2 + \sigma_S^2 w_S^2 + 2\sigma_R \sigma_S w_R w_S r}$$

$r = -1$

$$\sqrt{\sigma_R^2 w_R^2 + \sigma_S^2 w_S^2 - 2\sigma_R \sigma_S w_R w_S}$$

$\because \quad a^2 + b^2 - 2ab = (a-b)^2$

$\therefore$

$$\sigma_p = \sqrt{(\sigma_R w_R - \sigma_S w_S)^2}$$

$$= \sqrt{\left[(0.2)(0.8) - (0.5)(0.2)\right]^2}$$

$$= \sqrt{(0.06)^2}$$

$$= 0.06$$

$$= \boxed{6\%}$$

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(iii) Returns are perfectly positively correlated

$\therefore r = 1$

$$\therefore \sigma_p = \sqrt{\sigma_R^2 w_R^2 + \sigma_S^2 w_S^2 + 2 w_R w_S \sigma_R \sigma_S r_{RS}}$$

$$= \sqrt{\sigma_R^2 w_R^2 + \sigma_S^2 w_S^2 + 2 w_R w_S \sigma_R \sigma_S}$$

By applying, $a^2 + b^2 + 2ab = (a+b)^2$

$$\sigma_p = \sqrt{(\sigma_R w_R + \sigma_S w_S)^2}$$

$$\sigma_p = (0.2)(0.8) + (0.5)(0.2)$$

$$= 0.26$$

$$\sigma_p = \boxed{26\%}$$

## Answer-2(b)

Q.2.

b.

(i) Mr. X to buy ₹ 40,00,000 after 2 months

∴ Rate = ₹ 88.00 / $

$$\frac{40,00,000}{88}$$

= $ 45,454.55

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(ii) Mr. X to buy $ 300,000 Spot

∴ Rate = ₹ 86.20/$

= 300,000 × 86.2

= ₹ 25860,000.

(iii)

(a) If convert US $ now,

₹ equivalent = $75,000 × ₹86/$

= ₹64,50,000

Interest for 2 months = $64,50,000 \times \frac{2}{12} \times 12\%$

= ₹129,000

Total Proceeds after 2 months = 6450,000 + 129000

= ₹65,79,000

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(b) If convert in ₹ 2 months later

$75,000 × ₹88/$ = ₹66,00,000.

As Mr. X will receive ₹21,000 more (66,00,000 - 65,79,000) if converted US $ to ₹ 2 months later, he should convert <u>after 2 months</u>.

## Answer - 2(c)

Q. 2.
c.

In the post pandemic scenario, role of CFO has developed ~~befor~~. beyond traditional finance to also include:-

(i) Risk Management

CFO has to place more emphasis on risk management. in the post pandemic environment with various risks.

(ii) Supply Chain Financing

Global markets have been under turmoil since onset of pandemic. Supply chain activities have been affected due to these factors. These are inherently relevant from finance point of view. Hence, CFO has to play a crucial role.

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(iii) Mergers, Acquisitions and Corporate Restructuring

~~UJhi~~

With changing economic factors, corporate world has seen major mergers, acquisition and restructuring arrangements to inorginacally grow shareholder values. CFO plays a crucial role ~~be~~ to assess these arrangements as failure to carefully assess and analyze them may lead to major losses and even downfalls.

(iv) Environmental, Social and Governance Financing

Investor-focus has shifted from traditional parameters to new environmental, social and governence factors. Thus, focus of CFO has shifted from traditional finance to sustainable financing and growth.

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Answer - 1(a)

Q.1.

a. (i) $D_0 = 3.5$

$K_e = 12.5\%$

$g = 2\%$

$$P_0 = \frac{D_0(1+g)}{K_e - g}$$

$$= \frac{3.5(1.02)}{0.125 - 0.02}$$

$$= ₹34$$

(ii) P/E Multiple Approach

ROE = 8%

∴ P/E = ~~1/8 = 0.1~~

$$\frac{1}{0.08} = 12.5$$

EPS = 2.75

$P_0$ = EPS × P/E

= 2.75 × 12.5

$P_0$ = 34.375

Share Price = ₹45

∴ Share Price is <u>overvalued</u>.

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Earnings Growth Model

$$P_0 = \frac{EPS\,(1+g)}{ROE - g}$$

$$= \frac{2.75\,(1.02)}{0.08 - 0.02}$$

$$= ₹46.75$$

∴ Share Price is undervalued

Alternatively,

$$P_0 = \frac{2.75\,(1.02)}{0.125 - 0.02}$$

$$= ₹26.71$$

∴ Share Price is overvalued.

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## Answer - 1(b)

Q.1.

b. Financial leverage = 1.5 times

$$= \frac{EBIT}{EBT}$$

$$1.5 = \frac{EBIT}{EBIT - Interest}$$

$$1.5 = \frac{EBIT}{EBIT - (300\ lakh \times 8\%)}$$

$$1.5(EBIT - 24\ lakh) = EBIT$$

$$1.5\ EBIT - 36\ lakh = EBIT$$

$$0.5\ EBIT = 36\ lakh$$

$$\therefore EBIT = 72\ lakhs$$

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Total Equity = 100 + 200

= 300 lakhs

Debt = 300 lakhs

$\therefore$ WACC = $k_e w_e + k_d w_d$

= 15 (0.5) + 8 (0.75)(0.5)

= 7.5 + 3

= 10.5%

NOPAT = EBIT (1-t)

= 72 (1 - 0.25)

= 54 lakh

Economic Value Added = NOPAT - [WACC x Invested Capital]

= 54 - [600 lakh x 10.5%]

= 54 - 63

= -9 lakhs

[(9) lakhs]

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## Answer- 1 (c)

Q.1.

C. Following are the types of risk faced by Mr. PK :-

(i) Political Risk or Country Risk :-

Mr. PK is operating in more than one geography, it imports goods from China, processes in India and sells them in America. Thus, it will have to cater to laws and regulations of all three countries as per the operations undertaken by him. Further, negative economic effects can adversely affect its operations. Government actions like bans on Chinese imports or trade barriers by the American Government on exports from India could pose serious challenges for Mr. PK.

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(ii) Compliance Risk :-

Mr. PK needs to take care of various laws and regulations in all 3 countries, especially laws related to foreign trade, remittances and taxes on import and exports.

Non-compliance could lead to serious consequences including fines and penalties.

(iii) Foreign Exchange Risk :-

Import of raw materials and export of finished goods will require foreign exchange conversions and associated risks. Appreciation of Chinese Yuan and/or Depreciation of US Dollar could be detrimental to Mr PK's operations.

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(iv) Counter-Party Risk :-

Mr PK's transaction involves interaction with various parties. Failure on part of counter-party to fulfill its obligations could be ~~ne~~ detrimental.

Counterparty risk could either arise on non-delivery of goods by supplier for which payments have been made or non-receipt of consideration by customer for which goods have already been delivered.

(v) Other Risks could include :-

a) Strategic Risk - This arises when strategy fails to meet its objective.

b) Operational Risk :- This is risk from day-to-day transactions of entity. They can be people and/or process related risk.

c) Market Risk :- This could be risk due to various market factors like international prices ~~for~~ of raw material and finished goods.

d) Interest Rate Risk :- This risk arises due to mismatch between interest rates and also on account of changing market interest rates.

e) Liquidity Risk :- Liquidity risk arises when entity is unable to pay its obligations on due time.

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## Answer-3(a)

Q.3.

a.

(i) Initial Investment :-

| | |
|---|---|
| Cost of Machine | 70,00,000 |
| Working Capital | 10,00,000 |
| | 80,00,000 |

(ii) Before Tax Cash Operating Income in Real Terms :

₹ 25,00,000 p.a.

~~∴ After Tax Cash Flows =~~

Depreciation p.a. = $\frac{70,00,000}{5 \text{ years}}$

= ₹14,00,000

| | |
|---|---|
| Income Before Tax | 25,00,000 |
| (-) Depreciation | (14,00,000) |
| [It is assumed that depreciation is not included in the above Computation] | |
| | 11,00,000 |

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| | |
|---|---|
| Tax @ 35% | (385000) |
| After Tax Income | 715,000 |
| (+) Depreciation | 1400,000 |
| Cash Flows | 21,15,000 |

| Year | Cash Flow | PVF @10% | Present Value |
|---|---|---|---|
| 0 | (80,00,000) | ~~0.909~~ | 1 (80,00,000) |
| 1 | 21,15,000 | ~~0.826~~ | 0.909 1922535 |
| 2 | 21,15,000 | ~~0.751~~ | 0.826 1746990 |
| 3 | 21,15,000 | ~~0.~~ | 0.751 1588365 |
| 4 | 21,15,000 | | 0.683 1444545 |
| 5 | 21,15,000 | | 0.621 1313415 |
| 5 | 10,00,000* | | 0.621 621,000 |
| | | | 636,850 |

As NPV is positive, machine should be purchased.

* It is assumed that working capital is recovered at the end of year 5

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Alternatively,
If it is assumed that working capital is not recovered at end of Year 5,

NPV = 636,850 - 621,000

15,850

NPV is still positive,
project should be accepted.
ie. machine should be purchased

~~Eurt~~

Further,

Alternatively,

if it is assumed that depreciation of ₹14,00,000 is deducted to arrive at Before Tax Cash Income of ₹25,00,000 computation will be as follows:-

| | |
|---|---|
| Before Tax Cash Flow | 25,00,000 |
| (-) Tax @ 35% | (875000) |
| | 16,25,000 |
| (+) Depreciation | 1400,000 |
| Free Cash Flows | 30,25,000 |

| Year | Cash Flow | PVF@ 10% | Present Value |
|---|---|---|---|
| 0 | (80,00,000) | 1 | (80,00,000) |
| 1 | 30,25,000 | 0.909 | 27,49,725 |
| 2 | 3025000 | 0.826 | 24,98,650 |
| 3 | 3025000 | 0.751 | 22,71,775 |
| 4 | 3025000 | 0.683 | 20,66,075 |
| 5 | 3025000 | 0.621 | 18,78,525 |
| 5 | 10,00,000 | 0.621 | 621,000 |
| | | | ~~37,85,750~~ |
| | | | 40,85,750 |

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NPV is positive, machine should be purchased.

Alternatively, assuming working capital is not recovered at year 5,

NPV = 40,85,750 - 621,000

= 34,64,750

NPV is positive, machine should be purchased.

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## Answer - 3 (b)

Q.3.
b.

(i) Income to be Earned = 50,00,000 × 5%.

= 250,000

Cost to be recovered = 125,000

∴ Total Gross Income = 250,000 + 125,000

= ₹375,000

$$\text{Income} = 50,00,000 \times 9\% \times \frac{M}{12}$$

$$375000 = \frac{450,000\ M}{12}$$

$$M = 10$$

∴ Time = **10 months**

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(ii) Minimum Period to Break-Even.

Income to be Earned = Expenditure

$= 125,000$

$\therefore \quad 125,000 = 50,00,000 \times 9\% \times \frac{M}{12}$

$\therefore \quad M = 3.33$

$\therefore$ Minimum Time Period = 3.33 Months approx.

or

3 1/3 Months.

## Answer - 3(C)

Q.3.

C. Types of test to be performed for weak form of efficiency:-

(i) Serial Correlation Test

Under this method, price changes under two subsequent periods are taken into consideration. Say for example, price changes of Week 1 and Week 2 of a month.

Correlation between these subsequent periods shall be computed. Correlation generally tends to be very low, indicating weak form of market efficiency.

Multiple tests have been conducted on various time periods, showcasing weak form of efficiency.

However, such tests shall not be computed for longer periods as

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general price level always tends to be upwards

(ii) Runs Test :-

Price changes in market are treated as a run. One run shall include all consecutive price changes is one direction. On change in direction of change, another run shall begin

Example :-

++/---/++++/-/+++/----

1 2 3 4 5 6

These runs when computed on normal distribution basis shall indicate weak form of market efficiency.

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(iii) Filters Rule :-

Under this rule, person will invest ~~in~~ when it increases by N%.
It will hold it until a further N% fall in price.
On N% fall in price, share will be sold. This activity shall be conducted for a period, i.e. buy on N% rise, sell on N% fall.

When compared to buy and hold strategy, this test indicates weak form of market efficiency

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